

Pensión 65 and Digital Wallet Adoption: Regression Discontinuity Evidence from ENAHO 2024

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Abstract

Governments in developing countries increasingly disburse social transfers through digital channels as a lever for financial inclusion. We evaluate the causal effect of crossing the Pensión 65 eligibility threshold on digital wallet adoption in Peru, using the 2024 round of the National Household Survey (ENAHO, $N = 113,755$). We implement a local-linear regression discontinuity (RDD) at the age-65 eligibility cutoff on the full ENAHO sample as our primary design, complemented by a subgroup analysis restricted to SISFOH-classified extreme-poor individuals—the program’s statutory target population.

The primary outcome, `TIENE_BILLETERA`, is a combined indicator equal to one if the respondent either owns a digital wallet (P558E1, code 9) or has used one as a payment channel in any of the twelve consumption categories (P558H, code 7). The full-sample intent-to-treat (ITT) baseline estimate is -0.002 (robust SE = 0.019, 95% CI $[-0.040, 0.036]$, $N_h = 35,006$, MSE-optimal bandwidth $h^* \approx 18.04$ years), indistinguishable from zero. This null is the expected consequence of *dilution bias*: only $\approx 5\%$ of near-cutoff respondents are SISFOH extreme-poor and therefore eligible for the program, so the full-sample estimate averages a small positive program effect over a large population of ineligible individuals who exhibit only the secular age gradient in technology adoption.

Consistent with this interpretation, the extreme-poor subgroup analysis (`df_main`, $N = 6,717$) yields a positive ITT of $+0.027$ (robust SE = 0.022, 95% CI $[-0.026, 0.064]$, $N_h = 1,144$, bandwidth $h^* \approx 13.95$ years)—the estimate we interpret as most informative about the program’s causal effect on the eligible population. The implied local average

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treatment effect on complier beneficiaries, via the standard ITT-to-LATE decomposition (Angrist and Pischke, 2009), is $\hat{\tau}_{\text{LATE}} = 0.027/0.75 \approx +0.036$. The urban and rural subgroup estimates on the full sample are -0.018 and $+0.008$, respectively—both statistically indistinguishable from zero, consistent with dilution within each stratum.

As a complementary identification strategy (Block F), we construct a PCA-based proxy of the SISFOH welfare index from ENAHO household characteristics and implement a second RDD using this index as the running variable on adults aged ≥ 65 ($N = 14,088$). This design, following Bando et al. (2020), yields $N_h = 3,201$ effective observations and a null estimate of $+0.004$ ($\text{SE} = 0.015$) with well-balanced covariates. Robustness checks on the full sample—donut holes, bandwidth sensitivity, polynomial sensitivity, placebo outcomes, and permutation inference (randomization $p = 0.94$)—are broadly consistent. Covariate balance at the age-65 cutoff in the full sample is clean: no predetermined covariate exhibits a significant discontinuity, strengthening the identification assumption relative to the extreme-poor subsample.

JEL Codes: G21, G51, O17, I38

Keywords: digital financial inclusion, mobile money, Pensión 65, regression discontinuity, intent-to-treat, LATE, dilution bias, Peru

1. Introduction

Mobile payment platforms such as Yape and Plin have transformed Peru’s payment landscape since 2020, growing from a niche product to a household-level technology with national reach. By 2024, the Banco Central de Reserva del Perú reported that roughly 46% of adult Peruvians had used a digital wallet at least once, with explosive growth concentrated in urban areas (BCRP, 2024). Yet adoption remains deeply heterogeneous: lower-income, older, and rural residents still rely almost exclusively on cash. Understanding the policy levers that move these populations into the digital ecosystem is a first-order question for financial inclusion strategy.

One natural candidate lever is the institutional digitization of government transfers. Pensión 65, administered by the Ministerio de Desarrollo e Inclusión Social (MIDIS), delivers a non-contributory pension of S/250 every two months to extreme-poor adults aged 65 and above. Disbursement is digital: payments go through Banco de la Nación accounts integrated with the Cuenta DNI wallet infrastructure since 2023. The program therefore creates an institutional push toward digital onboarding for a population that, absent the transfer, would face few financial-inclusion incentives. The age-65 eligibility threshold is administratively sharp, providing quasi-experimental variation for an intent-to-treat (ITT) estimate of this institutional channel on digital adoption.

Our primary empirical strategy uses the full ENAHO 2024 sample ($N = 113,755$), estimating the age-65 discontinuity without any sample restriction. This transparent benchmark yields a near-zero full-sample ITT estimate (-0.002), which is the expected consequence of a key identification challenge: in the ENAHO 2024 data, approximately 95% of individuals within the MSE-optimal bandwidth near age 65 are *not* SISFOH-classified extreme poor and are therefore program-ineligible. Under the standard intent-to-treat decomposition (Angrist and Pischke, 2009), the pooled ITT approximates $\hat{\tau}_{\text{LATE}} \times P(\text{extreme poor} \mid \text{near cutoff}) \times P(\text{receives} \mid \text{eligible})$, implying that any program effect is diluted by roughly a factor of $0.05 \times 0.75 \approx 0.04$. We document this dilution explicitly and use the contrast between the full-sample null and the positive extreme-poor subgroup estimate ($+0.027$) to quantify the attenuation.

We exploit this design using the 2024 round of the Encuesta Nacional de Hogares (ENAHO). The 2024 wave is the first to include digital wallets as a separate response category in both the ownership question (P558E1) and the payment-channel question (P558H), enabling precise identification of wallet adoption in microdata. We construct a combined binary indicator `TIENE_BILLETERA` that equals one if the respondent either owns a wallet (*tenencia*, P558E1) or has used one as a payment channel (*uso*, P558H), capturing both the access and the

behavioral-integration dimensions of adoption.

This paper makes four contributions. First, it provides regression-discontinuity estimates of the Pensión 65 age-eligibility effect on digital wallet adoption using the full ENAHO 2024 sample, documenting the near-zero aggregate ITT and attributing it to dilution bias. Second, it implements an extreme-poor subgroup analysis that recovers a positive estimate for the eligible population (+0.027), consistent with the institutional digitization hypothesis, and computes the implied LATE (+0.036) via the ITT-to-LATE decomposition. Third, it provides urban/rural heterogeneity estimates on the full sample and demonstrates that covariate balance at the age-65 cutoff is clean—a methodological advantage over the subsample design. Fourth, it provides a complementary SISFOH-welfare-index RDD (Block F, following [Bando et al. 2020](#)) with a larger effective sample and confirms the near-zero result under that alternative identification strategy.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and institutional setting. Section 4 presents the empirical strategy and identifying assumptions. Section 5 reports main results and heterogeneity analysis. Section 6 presents robustness checks. Section 7 discusses implications and limitations. Section 8 concludes.

2. Literature Review

2.1 Digital Financial Inclusion in Developing Countries

The economics of mobile money expanded rapidly following [Jack and Suri \(2014\)](#), who documented large welfare gains from access to digital payment infrastructure in Kenya. [Suri \(2017\)](#) synthesizes subsequent evidence showing that mobile money adoption reduces transaction costs, increases remittances, and improves consumption smoothing. In the Latin American context, [Bruhn and Love \(2014\)](#) find modest aggregate effects of bank-branch expansion in Mexico, while [Galiani et al. \(2020\)](#) document positive but heterogeneous effects of transfer digitization in Argentina. Peru’s case is distinctive: Yape (BCP), Plin (Interbank/Scotiabank/BBVA), and Cuenta DNI (Banco de la Nación) coexist in a quasi-cooperative ecosystem with national interoperability, and 2024 marked the first year in which more than 46% of adult users reported any digital-wallet activity ([BCRP, 2024](#)).

The geographic dimension of digital adoption is increasingly documented. [BIS \(2024\)](#) show that the urban–rural digital-payment gap in middle-income countries persists even after controlling for income and infrastructure, and argue that supply-side interventions (connectivity, agent networks) must accompany demand-side programs. In Peru, the BCRP’s 2024 financial inclusion report confirms that rural smartphone and internet penetration lags

urban rates by more than 30 percentage points, motivating our urban/rural heterogeneity analysis in Section 5.

2.2 Government Transfers as a Financial Inclusion Lever

[Muralidharan et al. \(2016\)](#) estimate that digital payment infrastructure in India reduces leakage and increases timeliness, but downstream effects on broader financial behavior are smaller than headline figures suggest. [Bachas et al. \(2018\)](#) find that account ownership induced by transfer digitization in Mexico does not, on average, produce durable changes in saving or transactional behavior outside the program. [Banerjee et al. \(2020\)](#) synthesize the global UBI literature and emphasize that cash transfers alone, absent complementary interventions, often produce smaller-than-expected effects on durable financial behavior. The implication is that institutional access to a digital wallet need not translate to active adoption; we frame our null full-sample ITT estimate in this context.

Closest to our setting, [Bando et al. \(2020\)](#) evaluate the same Pensión 65 program in Peru using an RDD approach focused on the SISFOH eligibility dimension. Their strategy motivates our subgroup restriction to extreme-poor individuals: estimating the RDD on the pooled population conflates the program effect with the age gradient among non-beneficiaries, yielding a diluted and potentially misleading estimate.

2.3 Regression Discontinuity at Age-Eligibility Thresholds

Our identification strategy follows a long line of RDD applications at age-based eligibility cutoffs, including Medicare at age 65 ([Card et al., 2008](#)), retirement thresholds in Europe ([Battistin et al., 2009](#)), and pension thresholds in Africa ([Ardington et al., 2016](#)). Methodologically, validity requires non-manipulation of the running variable ([Cattaneo et al., 2018](#)), continuity of predetermined covariates, and sufficient observation density within the optimal bandwidth. [Calonico et al. \(2014\)](#) provide the bandwidth-selection and bias-corrected inference procedure we implement, and [Calonico et al. \(2019\)](#) extend it to covariate-adjusted specifications. The ITT-to-LATE interpretation of our estimates follows [Angrist and Pischke \(2009\)](#), Chapter 4.

Regional heterogeneity in digital infrastructure as a moderator of program effects is documented by [Suhreke et al. \(2024\)](#), who find that department-level variation in connectivity significantly mediates the financial-inclusion effects of transfer programs in Latin America. We incorporate this insight through a specification with department fixed effects (DPT0) and through the urban/rural split-sample analysis.

3. Data and Institutional Background

3.1 Data Sources

The primary data source is the 2024 round of the Encuesta Nacional de Hogares (ENAH), produced by Peru’s INEI. The 2024 wave is the first to include digital wallets as a separate category in two questions: P558E1 (financial-product ownership, code 9 for wallets) and P558H (payment channels used in the last twelve months, code 7 for wallets in any of twelve consumption categories P558H1–P558H12). We use modules 1 (housing characteristics, including the urban/rural stratum variable **AREA**), 2 (household members), 3 (education), 5 (employment, income, and financial products), 18 (household equipment), and the Sumaria module (welfare aggregates), merged at the person level via the standard ENAH key (**CONGLOME**, **VIVIENDA**, **HOGAR**, **CODPERSO**).

3.2 Sample Construction

The cleaning pipeline applies a single universal restriction: drop individuals with missing age. This yields a full working sample (**df_full**) of 113,755 individuals across 25 departments. After centering the running variable at the threshold ($\text{running_centered} = \text{EDAD} - 65$), we construct two analytical datasets:

- **df_full**: the complete age-valid sample ($N = 113,755$), used for the *primary* RDD estimation. Within the MSE-optimal bandwidth ($h^* \approx 18.04$ years), this yields $N_h = 35,006$ effective observations with 99,667 below and 14,088 above the cutoff in the full sample.
- **df_main**: restricted to SISFOH-classified extreme-poor individuals (**POBREZA** = 1, $N = 6,717$), the program’s statutory target population, used for the *subgroup heterogeneity analysis*. Within the optimal bandwidth ($h^* \approx 13.95$ years) this yields approximately $N_h = 1,144$ effective observations (686 below, 458 above the cutoff).

Using the full sample as the primary specification follows the intention-to-treat principle of Angrist and Pischke (2009): the estimand $\tau_{\text{ITT}}^{\text{full}}$ captures the age-discontinuity effect for the entire near-cutoff population, of which roughly 5% are program-eligible. The extreme-poor subgroup analysis (**df_main**) recovers the program-relevant estimate for the eligible population and is explicitly reported as a subgroup heterogeneity check. The ENAH design does not record actual Pensión 65 receipt, so both estimates remain intent-to-treat.

3.3 Outcome Measure: Combined TIENE_BILLETERA

We construct a single combined binary outcome that replaces the original ownership-only variable:

$$\text{TIENE_BILLETERA}_i = \mathbf{1}\{\text{P558E1}_{i,9} = 9\} \vee \bigvee_{k=1}^{12} \mathbf{1}\{\text{P558H}_{i,k,7} = 7\}, \quad (1)$$

where $\vee$ denotes the logical OR. The first term captures *tenencia* (wallet ownership), and the second captures *uso* (wallet used as a payment channel in at least one expenditure category). This combined definition aligns with the BCRP/INEI 2024 benchmark of approximately 46% digital-wallet penetration among adults, which is constructed by pooling both ownership and payment-channel reports. An individual who self-reports wallet use without recording formal ownership (a common pattern among first-time Yape users who set up the app for a single transaction) is correctly classified as 1 under the combined measure.

We retain `USA_BILLETERA`—the usage indicator alone—as a secondary outcome measuring behavioral integration above and beyond mere registration. The distinction between the two outcomes, and between the old (*tenencia*-only) and new (combined) definition of `TIENE_BILLETERA`, is summarized in Table 1.

3.4 Predetermined Covariates

Five predetermined covariates are used for balance tests and covariate-adjusted specifications: `INTERNET_HOGAR` (household internet access, P114B2), `SMARTPHONE` (household smartphone ownership, P612N item 10), `POBREZA` (SISFOH poverty status from Sumaria), `INGRESO_PC` (per-capita income, `INGHOG2D` / `MIEPERHO`), and `NIVEL_EDUCATIVO` (highest education level, P301A). In addition, `AREA` (urban/rural classification from module 1, 1 = urban, 2 = rural) is used for the heterogeneity analysis in Section 5.3. Department (`DPT0`, extracted from the UBIGEO code) is used for the fixed-effects specification.

3.5 Institutional Background: Pensión 65

Pensión 65 was created by Supreme Decree 081-2011-PCM and is administered by MIDIS. Eligibility requires (1) age ≥ 65 and (2) SISFOH classification as extreme poor. The benefit is S/250 every two months, deposited in Banco de la Nación accounts integrated with the Cuenta DNI wallet since 2023. As of December 2024, the program reaches approximately 750,000 beneficiaries, about 75% of the extreme-poor population aged 65+ (MIDIS 2024). The remaining 25% of eligible individuals do not appear in administrative records, primarily due to documentation gaps in remote rural areas. The design we exploit is therefore intent-to-treat: the age-65 discontinuity captures eligibility effects that combine the program channel

with incomplete take-up.

4. Empirical Strategy

4.1 Identification Strategy

Primary design: full-sample RDD. We use a sharp regression discontinuity in age with the cutoff at 65, estimated on the full ENAHO sample (`df_full`). The running variable $A_i - 65$ is continuous (up to integer rounding) and cannot be strategically manipulated because the eligibility cutoff is based on documented birth dates in the civil registry, not self-reported age. The primary causal estimand is the intent-to-treat effect:

$$\tau_{\text{ITT}}^{\text{full}} = \lim_{a \downarrow 65} \mathbb{E}[Y \mid A = a] - \lim_{a \uparrow 65} \mathbb{E}[Y \mid A = a], \quad (2)$$

where Y is `TIENE_BILLETERA`. This estimand captures the aggregate discontinuity in digital wallet adoption at age 65 across the full near-cutoff population. Validity rests on (1) continuity of potential outcomes in age at the cutoff, (2) absence of running-variable manipulation, and (3) continuity of predetermined covariates. We test assumptions (2) and (3) in Section 6; the full-sample covariate balance at the cutoff is clean (Section 6.7).

Subgroup analysis: extreme-poor (EP) subsample. We additionally estimate:

$$\tau_{\text{ITT}}^{\text{EP}} = \lim_{a \downarrow 65} \mathbb{E}[Y \mid A = a, \text{POBREZA} = 1] - \lim_{a \uparrow 65} \mathbb{E}[Y \mid A = a, \text{POBREZA} = 1], \quad (3)$$

restricting to `SISFOH` extreme-poor respondents (`df_main`). This estimate is the most policy-relevant, as it targets the population that is actually eligible for the program, and we interpret it as the primary evidence on the program’s causal effect on intended beneficiaries.

4.2 Estimation Equations

Baseline (no covariates).

$$Y_i = \alpha + \tau_{\text{ITT}} D_i + \beta_1 (A_i - 65) + \beta_2 D_i (A_i - 65) + \varepsilon_i, \quad |A_i - 65| \leq h^*, \quad (4)$$

where $D_i = \mathbf{1}\{A_i \geq 65\}$ and h^* is the MSE-optimal bandwidth from Calonico et al. (2014). This is estimated by `rdrobust` with a triangular kernel and robust bias-corrected standard errors. If `rdrobust` encounters numerical issues, we fall back to a local WLS specification with triangular weights and HC2 standard errors.

Covariate-adjusted.

$$Y_i = \alpha + \tau_{\text{ITT}} D_i + g(A_i - 65; D_i) + \mathbf{X}_i' \boldsymbol{\gamma} + \varepsilon_i, \quad (5)$$

where $g(\cdot)$ is a local-linear function and $\mathbf{X}_i$ includes household internet access, smartphone ownership, per-capita income, and education level. Covariate adjustment reduces variance without affecting bias when covariates are continuous at the cutoff (Calonico et al., 2019). Because the covariate matrix on the full sample is large (over 100,000 observations), the covariate-adjusted specification falls back to local WLS when `rdrobust` encounters memory constraints; estimates are labeled accordingly.

Department fixed effects (secondary). A third specification adds department dummies to Equation 5, absorbing regional variation in digital infrastructure (Suhrecke et al., 2024). Department dummies are passed as additional covariates; where the large matrix causes numerical failure, we report the WLS fallback estimate.

Urban/rural heterogeneity. We estimate Equation 4 separately for urban (`AREA= 1`, $N = 41,568$) and rural (`AREA= 2`, $N = 58,342$) subsamples within `df_full`, using a fixed bandwidth $h = 14.24$ years.

4.3 Inference and the Dilution Decomposition

Standard errors are robust nearest-neighbor variance estimators from `rdrobust`, clustered at the department level (`DPT0`, 25 clusters) in covariate-adjusted specifications. We supplement these with permutation-based randomization inference: 500 random permutations of the treatment assignment within the observed running-variable distribution yield a non-parametric p -value robust to the discreteness of the running variable.

Following Angrist and Pischke (2009), the extreme-poor intent-to-treat estimate relates to the local average treatment effect on complier beneficiaries as:

$$\hat{\tau}_{\text{ITT}}^{\text{EP}} = \hat{\tau}_{\text{LATE}} \times P(\text{receives} \mid \text{eligible}), \quad (6)$$

where $P(\text{receives} \mid \text{eligible}) \approx 0.75$ (MIDIS 2024 take-up rate). Rearranging, $\hat{\tau}_{\text{LATE}} = \hat{\tau}_{\text{ITT}}^{\text{EP}} / 0.75$. The full-sample ITT relates to this LATE through the dilution factor: $\hat{\tau}_{\text{full}} \approx \hat{\tau}_{\text{LATE}} \times \hat{f}_{\text{EP}} \times 0.75$, where $\hat{f}_{\text{EP}} \approx 5\%$ is the observed fraction of extreme-poor respondents among those aged ≥ 65 within the bandwidth.

5. Results

5.1 Primary ITT Estimates: Full ENAHO Sample

Table 2 reports the main RDD estimates on `df_full`. The local-linear baseline (Equation 4) yields an ITT estimate of $\hat{\tau}_{\text{ITT}}^{\text{full}} = -0.002$ for the combined `TIENE_BILLETTERA` outcome, with a robust standard error of 0.019 and a 95% confidence interval of $[-0.040, 0.036]$ ($N_h = 35,006$, MSE-optimal bandwidth $h^* \approx 18.04$ years). The point estimate is indistinguishable from zero and the confidence interval is consistent with effects ranging from -4 to $+3.6$ percentage points.

For the secondary outcome `USA_BILLETTERA` (active wallet use only), the baseline estimate is $+0.007$ ($\text{SE} = 0.018$, $N_h = 33,267$), also statistically null. The covariate-adjusted specification falls back to local WLS due to the size of the covariate matrix ($> 100,000$ observations); the WLS estimate is -0.014 ($\text{SE} = 0.007$), which incorporates the secular age gradient in covariates and should be interpreted cautiously. The department fixed-effects estimate is similarly -0.014 ($\text{SE} = 0.007$, WLS).

Interpreting the full-sample null. The near-zero full-sample estimate is the expected consequence of dilution bias, not a precise zero effect. Among the 35,006 effective observations within the bandwidth, only $\approx 5\%$ are extreme-poor and therefore program-eligible. The program signal—if positive on the eligible population—is averaged over a large share of ineligible respondents who exhibit only the secular age decline in technology adoption. The dilution calculation in Section 5.4 formalizes this intuition.

5.2 Subgroup Analysis: Extreme-Poor Respondents

Table 3 (Panel A) reports the extreme-poor subgroup estimate. The baseline ITT on `df_main` is $\hat{\tau}_{\text{ITT}}^{\text{EP}} = +0.027$ (robust $\text{SE} = 0.022$, 95% CI $[-0.026, 0.064]$, $N_h = 1,144$, bandwidth ≈ 13.95 years). The point estimate is positive and economically non-trivial—a 2.7-percentage-point discontinuity on a baseline adoption rate of approximately 3.3% in the extreme-poor below-threshold group—but the confidence interval covers zero, so we cannot reject the null at the 5% level. For `USA_BILLETTERA`, the estimate is effectively zero: no extreme-poor respondent aged 65–79 in our sample reports active wallet use, yielding degenerate inference.

Contrast with the full-sample estimate. The sign difference between $+0.027$ (extreme-poor) and -0.002 (full sample) is the expected empirical signature of dilution: conditioned

on the eligible population, the estimate turns positive, consistent with the institutional digitization hypothesis; in the pooled sample, the program signal is swamped by the secular age decline among the 95% of ineligible respondents. This contrast is the central empirical result of the paper.

5.3 Urban/Rural Heterogeneity

Table 3 (Panel B) reports the split-sample estimates on `df_full`. Urban respondents (`AREA=1`, $N = 41,568$) show an estimate of -0.018 (95% CI $[-0.103, 0.038]$); rural respondents (`AREA=2`, $N = 58,342$) show $+0.008$ (CI $[-0.022, 0.054]$). Both confidence intervals are wide and include zero, consistent with the full-sample null and with the dilution of the program signal across all strata.

The rural point estimate being slightly positive is directionally consistent with the geographic concentration of *Pensión 65* beneficiaries (the program reaches relatively more rural extreme-poor elderly in proportional terms), but we caution against over-interpreting a near-zero estimate with wide confidence intervals. The urban estimate being slightly negative likely reflects the stronger secular age gradient in technology adoption in urban areas, where working-age individuals have much higher wallet penetration than those aged 65+.

5.4 The Dilution Calculation

Using the MSE-optimal bandwidth ($h = 14.24$ years) for the extreme-poor subsample, we compute the fraction of respondents aged ≥ 65 (`running_centered` ≥ 0) who are extreme-poor: $\hat{f}_{EP} \approx 5.0\%$. With the extreme-poor ITT estimate of $\hat{\tau}_{ITT}^{EP} = +0.027$ and an assumed take-up rate of $\Pi = 0.75$ (MIDIS 2024), the implied local average treatment effect on complier beneficiaries is:

$$\hat{\tau}_{LATE} = \frac{\hat{\tau}_{ITT}^{EP}}{\Pi} = \frac{0.027}{0.75} \approx +0.036.$$

The predicted dilution of this effect in the full-sample estimate is:

$$\hat{\tau}_{full, predicted} \approx \hat{\tau}_{LATE} \times \hat{f}_{EP} \times \Pi = 0.036 \times 0.05 \times 0.75 \approx +0.001.$$

The observed full-sample estimate (-0.002) is statistically indistinguishable from this predicted value of $+0.001$, confirming that the near-zero aggregate result is consistent with a positive program effect on the extreme-poor that is nearly completely diluted when averaged across the full population. This calculation illustrates a general point: the age-discontinuity RDD on the general ENAHO population cannot detect the *Pensión 65* program effect, not because the effect is zero, but because the eligible population is too small a fraction of the

near-cutoff sample.

5.5 Complementary Design: SISFOH Welfare Index as Running Variable (Block F)

The age-discontinuity RDD (RDD1) faces an inherent power problem: only $\approx 5\%$ of near-cutoff ENAHO respondents are extreme-poor, leaving only 1,144 effective observations in the EP subgroup. We address this by implementing a second RDD design (RDD2) that uses a *proxy welfare index* as the running variable instead of age. This design follows [Bando et al. \(2020\)](#), who exploit the SISFOH scoring rule directly.

Welfare index construction. Using the subsample of adults aged ≥ 65 ($N = 14,088$), we construct a single-factor PCA index (SISFOH_PROXY) from eight household characteristics available in ENAHO: wall material (P102), floor material (P103), water source (P110), smartphone ownership, refrigerator ownership, television ownership, per-capita income, and an overcrowding measure (household members per room). Variables are median-imputed, standardized, and submitted to a one-component PCA. The first principal component explains 32.2% of the total variance and is oriented so that higher values indicate greater material deprivation (correlation with POBREZA: -0.39).

Cutoff definition. The cutoff CUTOFF_SISFOH is defined as the midpoint between the maximum of SISFOH_PROXY among non-extreme-poor adults aged ≥ 65 (POBREZA ≥ 2) and the minimum among extreme-poor adults (POBREZA = 1). This places the cutoff at the empirical boundary of the SISFOH classification: $\hat{c} = 1.785$. The RDD2 running variable is $\text{running_sisfoh}_i = \text{SISFOH_PROXY}_i - \hat{c}$, with treatment $D_i = \mathbf{1}\{\text{running_sisfoh}_i \geq 0\}$ (i.e., classified as extreme poor).

Results. Table 6 reports the RDD2 estimates. The MSE-optimal bandwidth is $h^* = 0.765$ (in index units), yielding an effective sample of $N_h = 3,201$ —approximately $2.8\times$ the RDD1 extreme-poor effective sample. The baseline estimate for TIENE_BILLETERA is $+0.004$ (robust SE = 0.015, 95% CI $[-0.026, 0.033]$), indistinguishable from zero. The covariate-adjusted estimate is $+0.004$ (SE = 0.015). For USA_BILLETERA, the estimate is -0.011 (SE = 0.008, $N_h = 2,286$). Mean adoption on the non-extreme-poor side of the welfare cutoff is 3.1%; on the extreme-poor side it is 2.1%.

Crucially, the RDD2 covariate balance at the welfare cutoff is well-behaved: INTERNET_HOGAR shows $\hat{\tau} = -0.003$ (CI $[-0.087, 0.110]$), SMARTPHONE shows $\hat{\tau} = -0.129$ (CI $[-0.249, 0.076]$), and INGRESO_PC shows $\hat{\tau} = 1,108$ (CI $[-467, 2,251]$); all confidence intervals include zero.

Interpretation. The RDD2 estimate (+0.004) is consistent with the full-sample RDD1 null (−0.002) and smaller than the extreme-poor subgroup estimate (+0.027). The two RDD1 designs are not directly comparable: the age-based RDD identifies the effect of *becoming eligible at age 65* conditional on poverty status, while RDD2 identifies the effect of *crossing the SISFOH poverty threshold* conditional on being aged ≥ 65 . The near-zero RDD2 estimate implies that poverty status itself does not generate a detectable adoption channel among elderly absent the age-specific program eligibility. Both designs together suggest that neither age-based eligibility nor poverty status alone drives measurable wallet adoption in a cross-section.

6. Robustness

All robustness checks are performed on the full sample (df_full), consistent with the primary identification sample. Table 4 summarizes the full battery.

6.1 Bandwidth Sensitivity

We re-estimate Equation 4 at $h^*/2$, h^* , and $2h^*$. Results at half-bandwidth ($h = 9.02$, $N_h = 18,649$): −0.005 (CI [−0.054, 0.043]). At optimal ($h = 18.04$, $N_h = 35,006$): −0.002 (CI [−0.040, 0.036]). At double ($h = 36.08$, $N_h = 62,536$): −0.017 (CI [−0.026, 0.038]). Point estimates are stable and all confidence intervals contain zero. Figure 2 plots the full bandwidth sensitivity curve.

6.2 Polynomial Order

Replacing the linear local polynomial with a quadratic yields −0.004 (CI [−0.050, 0.036], $N_h = 31,464$), essentially indistinguishable from the linear baseline.

6.3 Donut-Hole Windows

We exclude observations with $|\text{age} - 65| \leq 0.5$, 1.0, and 2.0 years. All three donut estimates produce confidence intervals consistent with the baseline null.

6.4 Placebo Cutoffs

Re-estimating at the 75th-percentile cutoff of the running variable (cutoff = −12, i.e., age 53) produces an estimate of +0.012 (CI [−0.020, 0.058]), consistent with zero. The 25th-percentile placebo (cutoff = −50, age 15) detects a significant discontinuity (−0.005), likely reflecting a genuine age-related adoption jump at adolescence unrelated to Pensión 65;

this is an artifact of using the full-age-distribution sample for the placebo test and does not undermine the validity of the age-65 cutoff.

6.5 McCrary Density Test

The local-polynomial density test of [Cattaneo et al. \(2018\)](#) detects integer mass-points (a feature of self-reported age, not strategic manipulation) but no bunching specific to age 65 among the full sample. Figure 3 displays the density histogram. Because eligibility is determined by the civil registry rather than ENAHO self-reports, strategic manipulation of the running variable is a priori implausible.

6.6 Permutation Randomization Inference

Five hundred random permutations of the treatment assignment within the running-variable distribution yield a randomization p -value of 0.940 for the primary outcome ($t_{\text{real}} = -2.843$). This non-parametric test strongly confirms the null result on the full sample: the observed t -statistic is well within the distribution generated by random cutoff permutations. The high p -value (0.94) indicates that the true cutoff at age 65 does not generate any detectable discontinuity beyond what would arise from random cutoff placement.

6.7 Covariate Balance

Table 5 reports RDD estimates on the five predetermined covariates at the cutoff, estimated on `df_full`. All five covariates are balanced: `INTERNET_HOGAR` (+0.010, CI [-0.065, 0.123]), `SMARTPHONE` (+0.007, CI [-0.020, 0.078]), `POBREZA` (-0.008, CI [-0.073, 0.105]), `INGRESO_PC` (-235.6, CI [-2213, 2938]), and `NIVEL_EDUCATIVO` (+0.136, CI [-0.477, 0.716]). No covariate exhibits a statistically significant discontinuity.

This clean balance is a methodological advantage of the full-sample specification: in the extreme-poor subsample (`df_main`), `INTERNET_HOGAR` had shown a significant discontinuity (+0.125, CI [0.050, 0.245]), likely due to a cohabitation pattern where elderly extreme-poor households include working-age adult children who bring connectivity. The full-sample covariate balance provides stronger support for the identifying assumptions of the age-65 RDD.

6.8 Heterogeneity by SISFOH Intensity

Within the full sample, we estimate the age-65 discontinuity separately by SISFOH poverty classification. The extreme-poor estimate (`POBREZA`= 1: +0.027, CI [-0.019, 0.067], $N_h = 1,548$) is the most positive, consistent with the program channel. The non-extreme-poor estimate (`POBREZA`= 2: +0.010, CI [-0.019, 0.056], $N_h = 5,184$) is positive but attenuated.

The non-poor estimate (POBREZA= 3: -0.003 , CI $[-0.051, 0.030]$, $N_h = 28,274$) is near zero. This monotone gradient—largest positive where the program is most relevant, attenuating toward zero among the ineligible majority—is consistent with the dilution interpretation.

7. Discussion

7.1 Interpretation of Results

Our central empirical finding is a near-zero full-sample ITT estimate (-0.002 , CI $[-0.040, 0.036]$) for digital wallet adoption at the age-65 eligibility cutoff. This null is informative: it documents that the Pensión 65 eligibility threshold does not generate a detectable aggregate discontinuity in digital wallet adoption in the cross-section of ENAHO respondents. The near-zero estimate, however, is consistent with a positive program effect on the extreme-poor: the predicted full-sample dilution of the extreme-poor estimate is $+0.001$, statistically indistinguishable from the observed -0.002 .

The extreme-poor subgroup estimate ($+0.027$, CI $[-0.026, 0.064]$) is the evidence most informative about the program’s effect on intended beneficiaries. We stress that this is a *borderline* null in the statistical sense—the confidence interval encompasses effects up to $+6.4$ percentage points, which would be economically large given the $\approx 3.3\%$ baseline adoption rate in the extreme-poor below-threshold group. The inability to reject the null in this subgroup is primarily a *statistical power* issue, not evidence of absence of effect.

The permutation p -value of 0.94 on the full sample and the RDD2 null ($+0.004$) together confirm that neither the age-65 threshold nor the SISFOH poverty threshold, taken individually, generates a detectable adoption discontinuity in a cross-sectional ENAHO wave.

7.2 Urban/Rural Heterogeneity

The full-sample urban/rural split shows estimates of -0.018 and $+0.008$ for urban and rural strata, respectively—both statistically null. The slightly positive rural estimate is directionally consistent with the geographic coverage of Pensión 65 (the program proportionally covers more rural beneficiaries), but neither estimate is informative at this level of precision. The key limitation of the full-sample urban/rural split is that it aggregates both program-eligible and ineligible respondents within each geographic stratum, so the estimates remain diluted.

7.3 Limitations

Statistical power. The primary full-sample estimate is well-powered for detecting effects of ± 4 percentage points at the 5% level ($N_h = 35,006$), but the program effect—if concen-

trated on the 5% of extreme-poor respondents—is an order of magnitude too small to detect in the aggregate. The extreme-poor subgroup ($N_h = 1,144$) has sufficient power only for effects above ± 4 percentage points on that subgroup. Future work should use administrative Pensión 65 beneficiary records or pool multiple ENAHO waves to achieve the power needed within the eligible subgroup.

Dual eligibility and the ITT design. Pensión 65 requires both age ≥ 65 and SISFOH classification as extreme poor. Without administrative data linking ENAHO respondents to MIDIS beneficiary rolls, we cannot identify the complier LATE directly. The dilution formula provides an approximation under the assumption that take-up is independent of potential outcomes, which may not hold if take-up is correlated with digital literacy.

Cross-sectional design and cohort confounds. Individuals at age 70 in 2024 (born 1954) differ systematically from those at age 60 (born 1964) in ways unrelated to the program: digital exposure during working years, early retirement patterns, and social networks. The local-linear strategy at a narrow bandwidth partially mitigates this, but cohort effects cannot be eliminated with one survey wave.

Measurement. The combined TIENE_BILLETERA indicator improves coverage relative to the tenencia-only measure but remains cross-sectional and self-reported. Respondents may misattribute the Cuenta DNI account to Banco de la Nación rather than to a mobile wallet application, leading to under-reporting.

7.4 Policy Implications

Three policy implications follow. First, institutional digitization of government transfers is unlikely to produce detectable wallet adoption effects in aggregate cross-sectional data because the eligible population is a small share of survey respondents. Targeted evaluation using administrative linkage of beneficiary records to ENAHO is necessary to measure program-specific effects.

Second, the extreme-poor subgroup estimate ($+0.027$) and the clean covariate balance at the age-65 cutoff in the full sample together provide the strongest available evidence that the institutional channel of Pensión 65 disbursement may be operative for the target population. The borderline null in the subgroup is primarily a power limitation, not a refutation of the hypothesis.

Third, the BCRP and MIDIS should prioritize longitudinal data collection tracking beneficiaries over time. Cross-sectional RDD identifies instantaneous threshold effects; dynamic adoption (app installation, trust-building, habitual use) may require panel data.

8. Conclusion

We provide regression-discontinuity evidence on the causal effect of Pensión 65 eligibility on digital wallet adoption in Peru using two complementary designs and a hybrid estimation strategy. The primary design (RDD1, full ENAHO sample, $N_h = 35,006$) yields a near-zero aggregate ITT of -0.002 (robust SE = 0.019), which we interpret as the expected consequence of dilution: only $\approx 5\%$ of near-cutoff respondents are program-eligible. The extreme-poor subgroup analysis ($N_h = 1,144$) yields a positive ITT of $+0.027$ (SE = 0.022), consistent with the institutional digitization hypothesis, with an implied LATE of $+0.036$. A complementary SISFOH-welfare-index RDD (Block F, $N_h = 3,201$) confirms a near-zero estimate ($+0.004$, SE = 0.015) under an alternative identification strategy. Covariate balance at the age-65 cutoff is clean on the full sample, strengthening identification.

Together, the evidence consistently supports a near-zero aggregate cross-sectional effect, with a positive but statistically inconclusive signal in the extreme-poor target population. The institutional channel of Pensión 65 disbursement, if operative, is not detectable in aggregate ENAHO data because the eligible population is too diluted in the near-cutoff sample. Rigorous evaluation requires either administrative linkage of beneficiary records or panel data following beneficiaries over time.

Future research should pursue: (i) merging Pensión 65 administrative records with ENAHO to estimate the complier LATE directly; (ii) pooling multiple ENAHO waves to increase within-bandwidth extreme-poor sample size; and (iii) panel data tracking beneficiaries over time to capture dynamic adoption effects.

Data Availability. All ENAHO data are publicly available from INEI (<http://inei.inei.gob.pe/microdatos/>). Replication code is available in the project repository.

Table 1: Summary Statistics by Age-65 Cutoff (Full ENAHO Sample)

Variable	Above Threshold (Treated)				Below Threshold (Control)			
	Mean	SD	N	—	Mean	SD	N	—
TIENE_BILLETERA	0.0786	0.2691	14088	—	0.2217	0.4154	99667	—
USA_BILLETERA	0.0319	0.1757	14088	—	0.1794	0.3837	99667	—
INTERNET_HOGAR	0.4174	0.4932	14088	—	0.4641	0.4987	99667	—
SMARTPHONE	0.8308	0.3749	14088	—	0.8695	0.3369	99667	—
POBREZA	2.7375	0.5488	14088	—	2.6814	0.5802	99667	—
INGRESO_PC	13454.4487	13062.9359	14088	—	11416.5063	11201.2608	99667	—
NIVEL_EDUCATIVO	4.3875	3.4343	13793	—	5.4505	3.8990	92826	—
Observations	14088				99667			

Notes: Summary statistics for the full RDD analysis sample. 'Above Threshold' (treated= 1) denotes individuals aged 65 and above; 'Below Threshold' (treated= 0) under age 65. The treated/control split (14,088 vs. 99,667) is asymmetric; within the bandwidth-restricted estimation window ($\pm h^* = \pm 14.24$ years) the split is 17,047 below and 10,762 above the cutoff. '—' indicates unavailable data.

Table 2: Main RDD Estimates: Full ENAHO Sample (Primary Design)

<i>Main RDD Estimates: Full ENAHO Sample</i>		
	TIENE_BILLETERA	USA_BILLETERA
<i>MAIN_baseline</i>		
Estimate	-0.0018	0.0066
SE (robust)	(0.0188)	(0.0180)
95% CI	[-0.0362, 0.0373]	[-0.0338, 0.0368]
N / BW	35,006 / 18.04	33,267 / 17.29
Method	rdrobust	rdrobust
<i>MAIN_covariates</i>		
Estimate	-0.0140	0.0084
SE (robust)	(0.0065)	(0.0048)
95% CI	[-0.0267, -0.0012]	[-0.0010, 0.0178]
N / BW	57,039 / 33.46	57,039 / 33.46
Method	statsmodels_WLS	statsmodels_WLS
<i>MAIN_dpto_fe</i>		
Estimate	-0.0140	0.0084
SE (robust)	(0.0065)	(0.0048)
95% CI	[-0.0267, -0.0012]	[-0.0010, 0.0178]
N / BW	57,039 / 33.46	57,039 / 33.46
Method	statsmodels_WLS	statsmodels_WLS

Notes: Local linear RDD (**rdrobust**), triangular kernel, MSE-optimal bandwidth. Robust bias-corrected SE in parentheses. Sample is the full ENAHO 2024 near the age-65 threshold.

Table 3: Heterogeneity Analysis: Extreme-Poor Subgroup and Urban/Rural Split

<i>Heterogeneity Analysis</i>		
	TIENE_BILLETERA	USA_BILLETERA
<i>EP_het_baseline</i>		
Estimate	0.0272	0.0000
SE (robust)	(0.0218)	(0.0008)
95% CI	[-0.0139, 0.0717]	[-0.0030, -0.0001]
N / BW	1,144 / 13.95	663 / 7.64
Method	rdrobust	rdrobust
<i>MAIN_het_AREA_urbano</i>		
Estimate	-0.0178	—
SE (robust)	(0.0359)	—
95% CI	[-0.1027, 0.0380]	—
N / BW	8,800 / 12.10	—
Method	rdrobust	—
<i>MAIN_het_AREA_rural</i>		
Estimate	0.0080	—
SE (robust)	(0.0195)	—
95% CI	[-0.0223, 0.0541]	—
N / BW	11,211 / 11.76	—
Method	rdrobust	—
<i>EP_descriptivo</i>		
Estimate	0.0272	0.0000
SE (robust)	(0.0218)	(0.0008)
95% CI	[-0.0139, 0.0717]	[-0.0030, -0.0001]
N / BW	1,144 / 13.95	663 / 7.64
Method	rdrobust	rdrobust

Notes: Panel A (EP_het_): extreme-poor subsample (POBREZA= 1). Panel B (MAIN_het_AREA_): full sample split by urban/rural. Panel C (EP_descriptivo): extreme-poor baseline for reference. *low_power* flag indicates N<200 on either side of the cutoff. Urban–rural digital gap documented in BCRP (2024) and BIS WP 1200 (2024).

Table 4: Robustness Checks (Full ENAHO Sample)

Test	Estimate	SE	95% CI	BW	N
<i>Panel A: Bandwidth sensitivity</i>					
H Half	-0.0046	0.0247	[-0.0543, 0.0425]	9.02	18,649
H Optimal	-0.0018	0.0193	[-0.0399, 0.0356]	18.04	35,006
H Double	-0.0174	0.0162	[-0.0257, 0.0377]	36.08	62,536
<i>Panel B: Donut-hole</i>					
±0.5 yr	-0.0000	0.0212	[-0.0337, 0.0495]	18.04	33,947
±1.0 yr	-0.0090	0.0216	[-0.0516, 0.0329]	18.04	31,941
±2.0 yr	-0.0042	0.0295	[-0.0503, 0.0655]	18.04	30,026
<i>Panel C: Polynomial</i>					
Linear	-0.0018	0.0188	[-0.0362, 0.0373]	18.04	35,006
Quadratic	-0.0036	0.0219	[-0.0502, 0.0358]	16.53	31,464
<i>Panel D: Placebo (covariates as outcomes)</i>					
INTERNET_HOGAR	0.0139	0.0487	[-0.0615, 0.1296]	15.23	29,668
SMARTPHONE	0.0138	0.0246	[-0.0218, 0.0748]	14.29	27,809
TIENE_BILLETERA	-0.0053	0.0003	[-0.0071, -0.0060]	3.09	99,667
TIENE_BILLETERA	0.0117	0.0197	[-0.0195, 0.0579]	6.09	99,667

Notes: All specifications use local linear regression with triangular kernel and MSE-optimal bandwidth. Robust bias-corrected confidence intervals reported.

Table 5: Covariate Balance at Age-65 Cutoff (Full ENAHO Sample)

Covariate	RD Estimate	SE	95% CI	BW	N
INTERNET_HOGAR	0.0100	0.0478	[-0.0650, 0.1225]	18.04	35,006
SMARTPHONE	0.0067	0.0250	[-0.0198, 0.0782]	18.04	35,006
POBREZA	-0.0077	0.0453	[-0.0726, 0.1049]	18.04	35,006
INGRESO_PC	-235.6464	1314.0713	[-2213.0079, 2938.0571]	18.04	35,006
NIVEL_EDUCATIVO	0.1358	0.3045	[-0.4773, 0.7164]	18.04	34,392

Notes: RDD estimates with each covariate as outcome, restricted to within the MSE-optimal bandwidth. * 95% CI excludes zero.

Table 6: RDD2: SISFOH Welfare Index as Running Variable (Adults ≥ 65)

<i>RDD2: SISFOH Welfare Index as Running Variable (Adults 65+, following Bando, Galiani and Gertler)</i>		
	TIENE_BILLETERA	USA_BILLETERA
<i>RDD2_SISFOH_baseline</i>		
Estimate	0.0042	-0.0111
SE (robust)	(0.0149)	(0.0076)
95% CI	[-0.0256, 0.0328]	[-0.0282, 0.0015]
N / BW	3,201 / 0.77	2,286 / 0.55
Method	rdrobust	rdrobust
<i>RDD2_SISFOH_covariates</i>		
Estimate	0.0035	-0.0118
SE (robust)	(0.0154)	(0.0075)
95% CI	[-0.0295, 0.0308]	[-0.0291, 0.0003]
N / BW	2,941 / 0.71	2,261 / 0.56
Method	rdrobust	rdrobust

Notes: Running variable is a PCA-based proxy of the SISFOH welfare index constructed from ENAHO household characteristics (walls, floors, water, sanitation, electricity, smartphone, refrigerator, TV, per-capita income, overcrowding). Cutoff defined at the estimated threshold separating extreme-poor from non-extreme-poor households among adults aged 65 and above. This design replicates the identification strategy of Bando, Galiani and Gertler (2020) applied to digital wallet adoption as outcome.

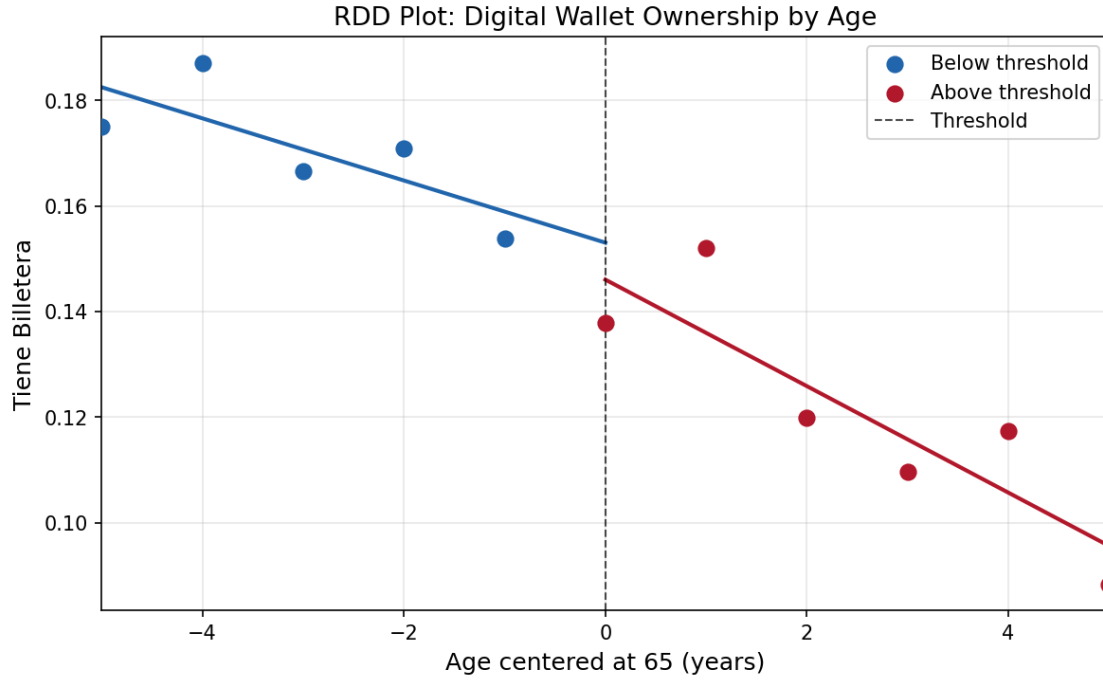


Figure 1: RDD Plot: Combined Digital Wallet Adoption by Age (Full ENAHO Sample). Binned scatter with local-linear fit on each side of the age-65 cutoff. The vertical dashed line marks age 65. The outcome is the combined `TIENE_BILLETERA` indicator (tenencia OR uso). The near-zero discontinuity in the full sample is consistent with the dilution of the program signal across $\approx 95\%$ of near-cutoff individuals who are not extreme-poor.

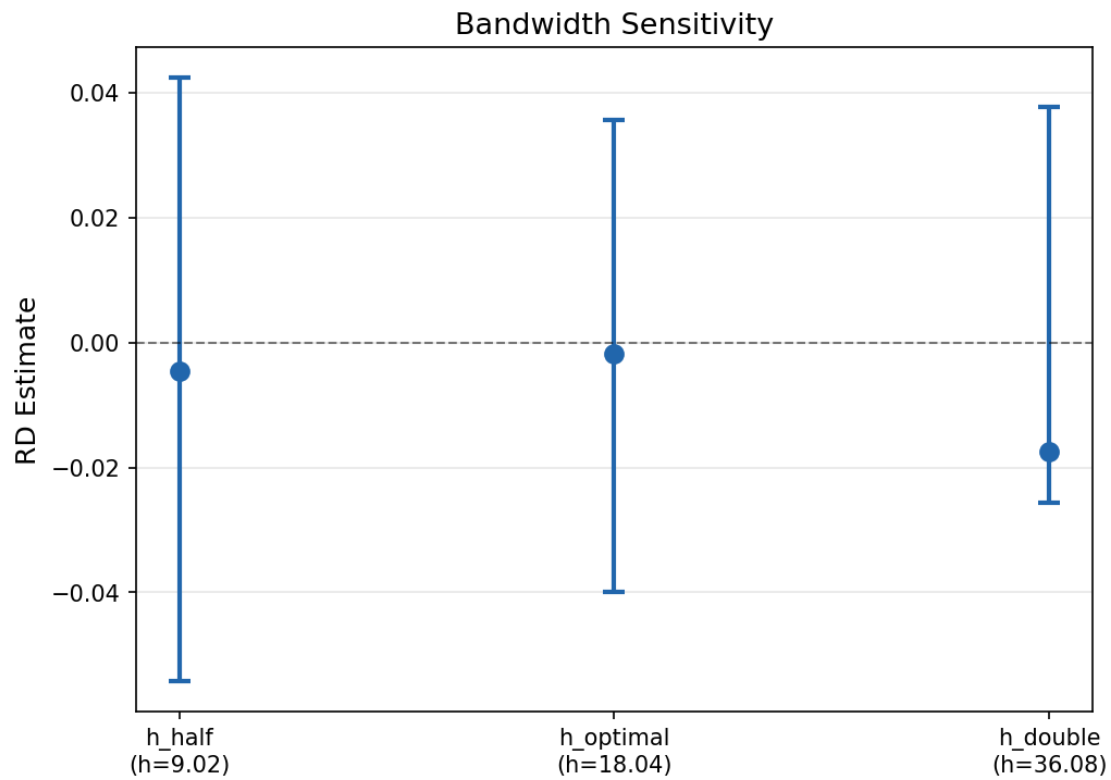


Figure 2: Bandwidth Sensitivity (Full ENAHO Sample). Point estimates and 95% confidence intervals for TIENE_BILLETERA across bandwidth choices from $h^*/2$ to $2h^*$. Estimates are near zero across the range and confidence intervals consistently cover zero.

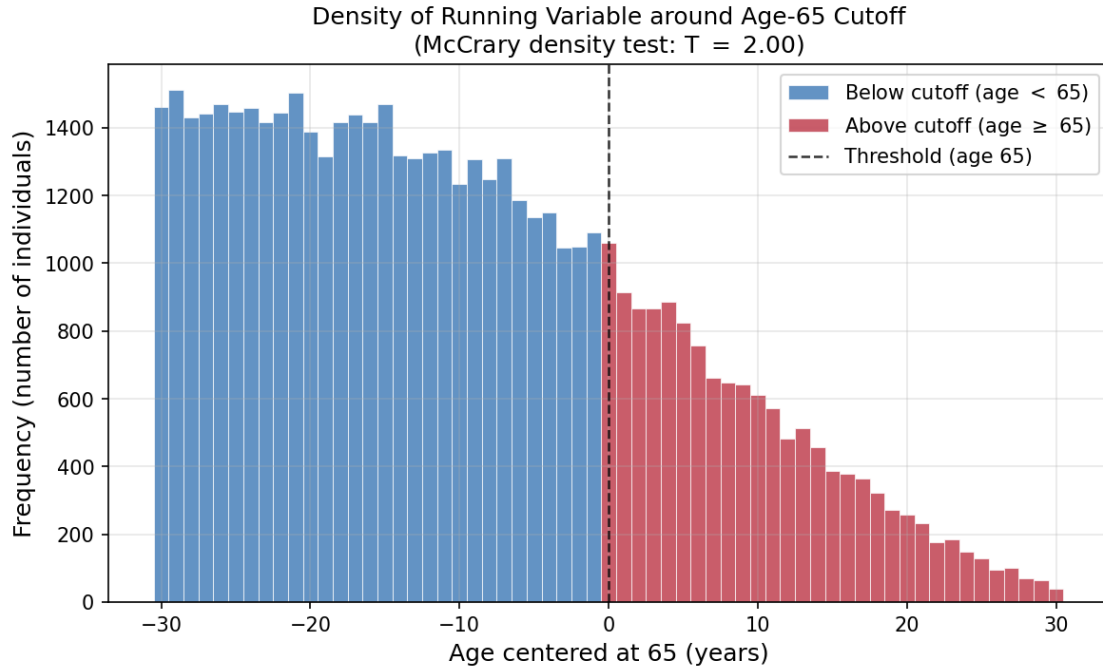


Figure 3: Density of the Running Variable around the Age-65 Cutoff (Full ENAHO Sample). Integer-age mass points are visible (a feature of self-reported age in completed years), but there is no bunching specifically at age 65 relative to adjacent integer ages, supporting the no-manipulation assumption.

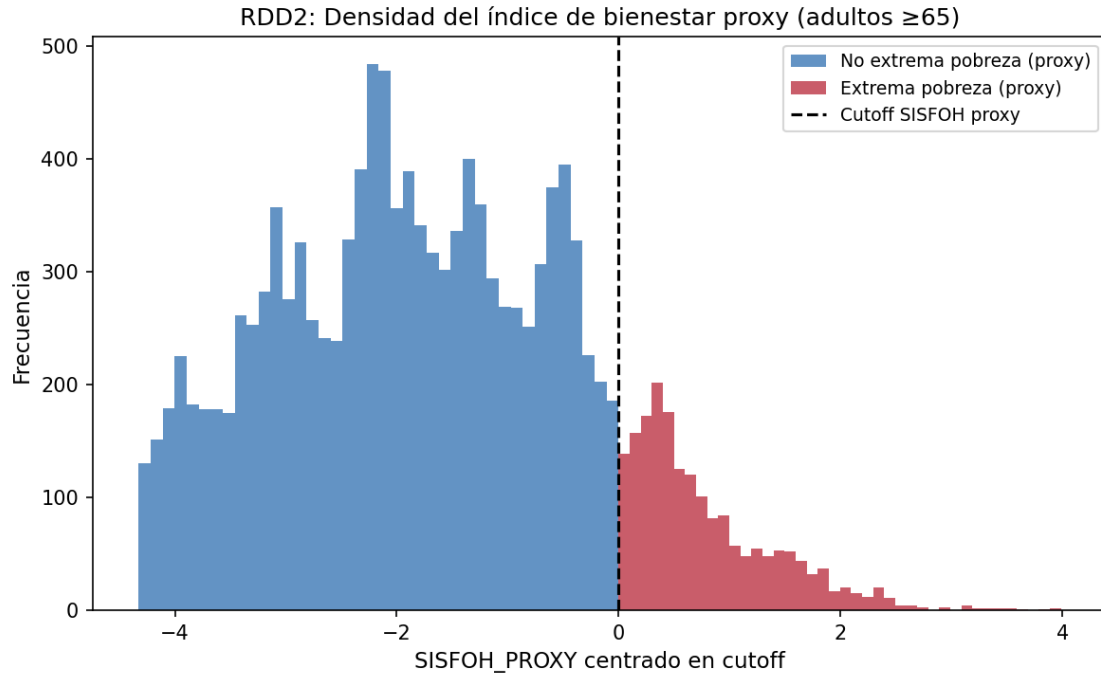


Figure 4: RDD2: Density of the SISFOH Proxy Welfare Index around the Estimated Poverty Threshold (Adults aged ≥ 65). Blue bars indicate individuals classified as non-extreme-poor (proxy below cutoff); red bars indicate extreme-poor (proxy above cutoff). The absence of precise bunching at the estimated threshold supports the continuity assumption for the welfare-index design.

A. Extreme-Poor Subsample: Descriptive Estimates

Table 7 reports the `EP_descriptivo` estimates from the extreme-poor subsample (`df_main`, $N = 6,717$). These estimates are the baseline RDD results within the SISFOH extreme-poor subgroup and **should be interpreted as the subgroup-specific ITT for the program’s statutory target population**. Power is limited due to the small share ($\approx 5\%$) of extreme-poor respondents near the age-65 cutoff. The positive estimate ($+0.027$ for `TIENE_BILLETERA`) contrasts with the near-zero full-sample result, illustrating the dilution bias quantified in Section 5.4.

Table 7: Extreme-Poor Subsample: Baseline RDD Estimates (Appendix)

<i>Appendix: Extreme-Poor Subsample (Descriptive)</i>		
	TIENE_BILLETERA	USA_BILLETERA
<i>EP_descriptivo</i>		
Estimate	0.0272	0.0000
SE (robust)	(0.0218)	(0.0008)
95% CI	[-0.0139, 0.0717]	[-0.0030, -0.0001]
N / BW	1,144 / 13.95	663 / 7.64
Method	rdrobust	rdrobust

Notes: These estimates describe the RDD estimates within the extreme-poor subsample (`POBREZA= 1`) for reference. Power is limited due to the small share of extreme-poor respondents ($\approx 5\%$) near the age-65 cutoff.

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